

Dataset for countries assessment regarding healthcare resilience

Datasets used in the paper <https://doi.org/10.1016/j.engappai.2025.113662>

Criteria that build model assessment cover 13 indexes presented in Table 1, where $\uparrow$ defines criteria with the aim of maximization, and $\downarrow$ denotes criteria with the aim of minimization.

Table 1: Multi-criteria model for healthcare evaluation.

Criteria group	Proposed indicators
G_1 - System's financial resources	C_1 - Public expenditure on health - share of GDP ($\uparrow$), C_2 - Public expenditure on health per capita ($\uparrow$)
G_2 - System's human resources	C_3 - Practicing physicians - density per 1 000 population ($\uparrow$), C_4 - Practicing nurses - density per 1 000 population ($\uparrow$)
G_3 - System's infrastructure	C_5 - hospital beds per 1 000 population ($\uparrow$), C_6 - curative (acute) care beds per 1 000 population ($\uparrow$)
G_4 - Service delivery	C_7 - Hospitalized patients per million ($\downarrow$), C_8 - Intensive Care Unit patients per million ($\downarrow$)
G_5 - Absorption of new medical products and technologies	C_9 - Total tests per thousand population ($\uparrow$), C_{10} - Vaccination rate (% of population vaccinated) ($\uparrow$)
G_6 - System's efficiency	C_{11} - Observed case-fatality ratio due to COVID-19 ($\downarrow$), C_{12} - Deaths per 1.000,000 population due to COVID-19 ($\downarrow$)
G_7 - System's robustness	C_{13} - Excess deaths (% change from average) ($\downarrow$)

Data was collected from the Our World in Data and OECD Health Statistics 2022 databases. The data is available in XLSX and CSV files.

Source of data:

Our World in Data: <https://ourworldindata.org/>

OECD Health Statistics 2022: <https://www.oecd.org/els/health-systems/health-data.htm>

The dataset consisting of performance values obtained from the mentioned data sources for 26 countries, regarding 25 European countries and the USA for 2021, is provided in Table 2.

Table 2: Dataset with performance values in relation to healthcare resilience evaluation collected for 2021.

Country	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8	C_9	C_{10}	C_{11}	C_{12}	C_{13}
United States	18.816	12196.97	2.67	12.5	2.77	2.53	58556.54	15168.83	1386.823	72.032	0.014147	1417.123	0.176779
Austria	9.555	5262.371	5.45	10.8	7.37	5.31	52604.16	11868.07	13234	74.842	0.010416	1063.525	0.092108
Belgium	11.065	6022.253	3.25	20.5	5.64	4.93	50511.13	12948.27	1743.986	76.689	0.006281	767.982	0.03289
Bulgaria	8.52	2479.297	4.168	4.74	7.454	5.95	36456.89	3342.987	904.108	28.805	0.04336	3453.262	0.393325
Czechia	9.24	4302.98	4.26	8.95	6.63	4.04	112347	20646.68	4084.551	64.951	0.013991	2355.455	0.216567
Denmark	9.264	5453.023	4.38	10.24	2.5	2.47	15783.19	2590.29	7309.683	80.118	0.003335	332.065	0.031456
Finland	9.613	5251.501	4.33	18.92	3.28	2.61	8934.355	1580.66	1119.22	77.435	0.005063	196.983	0.047004
France	12.205	6105.608	3.36	9.65	5.98	3	88607.16	14475.36	2328.867	78.371	0.008316	881.3	0.05194
Germany	11.016	6350.865	4.53	12.03	8	5.95	218595	12105.3	688.914	75.06	0.014924	961.138	0.069396
Greece	9.508	2736.445	6.306	3.77	4.21	3.64	4200	2491.2	4082.539	71.596	0.016135	1530.678	0.149173
Hungary	5.303	2748.683	3.28	6.59	7.02	4.23	132247.1	3045.9	667.733	61.548	0.031873	2971.14	0.194288
Ireland	5.275	4575.532	4.05	12.8	2.96	2.69	32018.62	5650.88	1521.554	78.248	0.005653	734.343	0.077525
Italy	7.145	3051.572	4.13	6.26	3.18	2.66	74866.85	8341.53	1872.367	81.786	0.017172	1090.165	0.105252
Lithuania	5.507	2320.756	4.47	7.88	6.56	5.06	90120.64	4628.2	1886.254	69.394	0.014869	2066.396	0.306496
Luxembourg	5.354	6273.571	2.98	12.1	4.51	3.22	32012	8089.836	3405.397	71.348	0.007876	668.859	-0.01734
Netherlands	9.551	5752.462	3.836	11.3	3.32	2.62	22918.16	8965.263	906.985	71.717	0.004176	553.548	0.097002
Norway	8.625	6044.488	5.18	18.37	3.6	3.1	8142.857	2846.9	1206.456	78.94	0.002532	156.153	0.036175
Poland	4.785	1863.608	3.44	5.68	6.62	4.37	101791.9	3240.32	528.472	54.167	0.024186	1701.611	0.22325
Portugal	7.171	2441.07	5.615	7.44	3.39	3.34	44826.74	7614.164	1971.867	91.636	0.013306	1189.633	0.082465
Romania	6.272	2384.52	2.974	7.37	6.892	4.03	112978.4	2316.769	630.598	30.503	0.036536	2201.216	0.2858
Slovakia	7.232	2521.59	3.68	5.73	5.82	4.85	106567.7	17871.67	7948.748	44.756	0.013184	2585.799	0.344704
Slovenia	6.828	2787.512	3.34	10.49	4.5	4.13	88757.99	19829.31	661.804	58.576	0.008661	1398.485	0.101913
Spain	10.714	4086.783	4.49	6.34	2.97	2.51	50710.18	12992.13	1048.113	83.424	0.009306	820.071	0.063362
Sweden	9.787	5358.066	4.32	10.85	2.22	1.9	31702.9	5318.717	1043.652	72.624	0.007819	652.269	0.025858
Switzerland	11.796	7582.184	4.45	18.39	4.53	3.59	35607.92	7751.984	1261.003	68.158	0.005351	513.403	0.050554
United Kingdom	9.898	4466.127	3.18	8.68	2.54	2.43	49970.98	5875.549	5061.414	74.079	0.008346	1247.407	0.090292